

UF HEALTH SHANDS HOSPITAL
DEPT OF PHARMACY SERVICESPOLICY NUMBER: **06-05-08**
CATEGORY: SATELLITE OPERATIONSORIGINATION DATE: 09/09/2010
LAST DATE UPDATED: 01/30/26
APPROVAL DATE: 02/13/2026TITLE: **Erythropoiesis-Stimulating Agents (epoetin alfa, or darbepoetin alfa)**

POLICY:

Due to the risk of thrombotic events when hemoglobin increases too rapidly or exceeds normal physiologic thresholds, Shands pharmacists will change doses, alter dosing times/days, order laboratory tests, discontinue and hold doses subsequent to erythropoiesis-stimulating agent (epoetin alfa or darbepoetin alfa) orders in accordance with this policy.

To support product stewardship, all subcutaneous doses of erythropoietin will be prepared in the pharmacy IV Center to minimize partial vial waste. Doses will be dispensed in luer lock syringes unless the dose of erythropoietin is the exact size of the vial.

PROCEDURE:

1. Upon receipt of the erythropoiesis-stimulating agent (ESA) order for patients greater than 1 year of age, the pharmacist will review the patient's hemoglobin and enter a one-time dose (standardized as per appendix A) if the hemoglobin is less than 10 g/dL for anemia associated with low-risk Myelodysplastic Syndrome, and 11.5 g/dL for anemia of chronic kidney disease.
 - a. If there is not a recent (within 72 hours) hemoglobin lab value available, the pharmacist may order the lab to determine if the patient meets laboratory criteria for use of the ESA therapy.
 - b. This information will be documented as a "therapeutic monitoring" I-vent in Epic
2. The initial dose should be dispensed if the patient's hemoglobin (Hgb) is less than 10 g/dL for anemia associated with low-risk Myelodysplastic Syndrome or less than 11.5 g/dL for anemia of chronic kidney disease.
 - a. Initial dose/drug will be standardized as per appendix A for all adult patients and any pediatric patients greater than or equal to 15 kg
 - b. Any subsequent dose must be approved and will be standardized for administration on either Monday or Thursday (see appendix B).
 - c. Approval will be based on laboratory data including: Hgb, iron studies, B-12 and folate levels (see appendix C).
 - d. Iron or other supplementation will be discussed with the prescriber when necessary (see appendix D).
 - e. Dose adjustments will be made as per appendix E.
 - f. ESA therapy is to be discontinued under the conditions listed in Appendix F.
3. For orders of epoetin alfa where the prescriber instructs that no substitution is allowed, dose preparation guidelines are as follows:
 - a. Doses < 200 units, the 500 units/mL dilution should be used
 - b. Doses ≥ 200 units and less than 1,000 units, the 2,000 units/mL concentration should be used
 - c. Doses of 1,000 units up to and including 10,000 units, the 10,000 unit/mL concentration should be used

- d. Doses > 10,000 units, the 40,000 units/mL concentration should be used.
 - e. The minimum dosing increment is 50 units (i.e., 250, 300, 350, etc.)
4. Subsequent doses require evaluation by a pharmacist and will be standardized as per appendix B.
 - a. If the patient does not meet criteria for use, the medication can be either discontinued or put on hold by the pharmacist.
 - b. If the patient does not meet criteria secondary to the need for iron, B-12 or folate supplementation, the pharmacist will discuss the deficiency with a provider from the patient's primary service.
 - c. Dose adjustments secondary to low response or higher than goal increases will be made as per appendix E with dose standardization as per appendix A.
 5. All recommendations (dose adjustments, ordering of laboratory tests, iron, folate or B-12 supplementation, etc.) and alterations to the ESA regimen will be documented in both the progress note section of the patient's chart and updated in the iVent linked to that patient's ESA order.
 6. Chart documentation of all modifications to the original order will be completed via placement of a P&T authorized order and a progress note signed by a pharmacist.
 7. Recommendations for the management of anemia in Jehovah's Witness patients or patients refusing blood transfusions can be found in Appendix G. Other special populations (ex. oncology patient with chronic kidney disease) will be addressed on a case-by-case basis. Therapy and goals of therapy will be determined jointly by the prescriber and pharmacist. Therapy and outcomes will be tracked with the intention of incorporating the needs of special populations into the next revision of the ESA policy.

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If you need to edit this policy and would like a version in Word, contact the Pharmacy Administrative Supervisor.

Appendix A: Dose Standardization

Depending on the formulary preferred agent at the time of the order, epoetin alfa and darbepoetin alfa can be freely substituted for each other unless the prescriber specifies "Do Not Substitute" on the original order. All* doses will be changed to an equivalent weekly dose as per the following charts:

Epoetin to darbepoetin dose adjustment chart

<u>Epoetin alfa (units / week)</u>	<u>Darbepoetin alfa (mcg / week)</u>
Less than 2500	6.25
2500-4999	12.5
5000-10,999	25
11,000-17,999	40
18,000-33,999	60
34,000-89,999	100
Greater than 90,000	100

Conversion from darbepoetin alfa to epoetin alfa to be accomplished via the following formula:

Step 1: If every-other week dosing, divide by 2

Step 2: Use dose ratio of 250 units of epoetin alfa to 1 mcg of darbepoetin alfa to calculate conversion

Step 3: Round dose up to the nearest vial/syringe if the increase in dose is less than 10% of the calculated dose. All other doses will be round down to the nearest vial/syringe size.

*Note: If epoetin alfa is the formulary preferred agent at the time of the order, three-times a week dosing orders for epoetin alfa do not have to be changed to weekly doses; they can remain as three-times-a week doses.

*Note: Pediatric patients will receive appropriate weight-based dosing of epoetin alfa until data supports the use of darbepoetin alfa in this population.

Appendix B: Day of Administration Standardization

<u>Day of First ESA Order</u>	<u>Day of Next Weekly Dose</u>
Monday	Monday
Tuesday	Monday of the following week (6 days)
Wednesday	Thursday of the following week (8 days)
Thursday	Thursday
Friday	Thursday of the following week (6 days)
Saturday	Monday of the following week (9 days)
Sunday	Monday of the following week (8 days)

*Note: If epoetin alfa is the formulary preferred agent at the time of the order, three-time-a-week orders for epoetin alfa do not have to be changed to weekly doses; they can remain as three-times-a-week doses.

Appendix C: Laboratory Monitoring

This policy allows the pharmacist to order a variety of laboratory tests. All relevant laboratory results will be reviewed before additional tests are ordered. The goal for laboratory monitoring is to address various causes of anemia and decrease unnecessary or potentially risky administration of ESAs.

- **Complete blood count:** Order if there is not a result in the previous week.
- **Hemoglobin:** Order if there is not a result in the previous 72 hours
- **Folic acid:** Order this test if MCV or MCH is elevated. If low, discuss with patient's primary team (see Appendix D for supplementation suggestions). Some medications cause macrocytic anemia. It is warranted to check levels once if such medications are used and folic acid deficiency is still suspected.
 - Azathioprine
 - Cyclophosphamide
 - Methotrexate
 - Oral contraceptives
 - Phenobarbital
 - Phenytoin
 - Sulfasalazine
- **Hemoglobin:** Order weekly on standard day of administration if not already ordered. Use to adjust dose as per Appendix E.
- **Iron studies:** Order serum ferritin, serum iron (SI), and total iron-binding capacity (TIBC) if either Mean Cell Volume (MCV) or Mean Cell Hemoglobin (MCH) are low. If ferritin, serum iron and iron saturation are low, and TIBC is increased, discuss iron supplementation with the primary team (see Appendix D for suggestions). One week after last dose of IV iron recheck iron studies; then order follow-up studies every 1 month until stable. If patient has hospital length of stay exceeding 20 days, order iron studies if not previously ordered.
- **Mean Cell Hemoglobin (MCH):** If elevated, evaluate for folate deficiency. If low, evaluate for iron deficiency (serum ferritin, serum iron, and total iron-binding capacity).
- **Mean Cell Volume (MCV):** If elevated (macrocytic) evaluate for vitamin B-12 and folate deficiency. If B-12 and folate are normal, review the current medications for possible drug-induced bone marrow effects; possible medications include: mercaptopurine, fluorouracil, cyclophosphamide, methotrexate, hydroxyurea, and zidovudine. If MCV is low, (microcytic) evaluate for iron deficiency (serum ferritin, serum iron, and total iron-binding capacity).
- **Reticulocyte Count:** Patients with untreated anemia (secondary to iron, folate or B-12 deficiency) will not see increased reticulocyte count. Appropriate supplementation should result in increase in reticulocyte count over 5-7 days. Order this lab to evaluate response to supplementation.
- **Vitamin B-12 (cyanocobalamin):** Order this test if MCV is elevated. If low, discuss with the patient's primary team (see Appendix D for supplementation suggestions). Some medications cause macrocytic anemia (see folic acid). It is warranted to check levels once if such medications are used and B-12 deficiency is still suspected.

Lab Name	Normal	If High	If Low
MCV, cu microns	78-100	Order vitamin B-12 & folate levels to evaluate for deficiency	Order serum iron, ferritin, and TIBC studies
MCH, pg	26-34	Order folate level to evaluate for deficiency	Order serum iron, ferritin, and TIBC studies
Vitamin B-12, pg/mL	243-894	--	Discuss initiating or optimizing cyanocobalamin supplementation
Folate, ng/mL	4.4-19.9	--	Discuss initiating or optimizing folic acid supplementation
Serum Iron, mcg/dL	45-160	--	Discuss initiating or optimizing iron supplementation
Serum Ferritin, ng/mL	25-400	--	Discuss initiating or optimizing iron supplementation
TIBC, mcg/dL	225-430	Discuss initiating or optimizing iron supplementation	---
Iron Saturation, %	20-55	--	Discuss initiating or optimizing iron supplementation

Appendix D: Supplementation Suggestions/Options for Iron, Folic Acid, Vitamin B-12

1. Iron

- The National Kidney Foundation recommends supplemental iron to maintain transferrin saturation (TSAT) of at least 20% and a serum ferritin of at least 100 ng/mL.
- Evaluate for infectious process and consider balance between infection risk (potential increase with administration of iron) and need for supplemental iron.
- Iron supplementation should be used with caution in patients with serum ferritin greater than 1000 mcg/L or with transferrin saturation greater than 50% due to possible toxicity.
- Consider IV iron if patient requires proton pump inhibitor therapy.
- Consider IV iron if oral iron does not improve ferritin or if oral iron is not well tolerated.
- Evaluate for potential interactions and time administration of iron appropriately.
- Patients who fail solid oral dosage forms of iron may respond well to ferrous sulfate liquid.
- Supplemental iron therapy is recommended for hemodialysis patients when the TSAT is less than 20%, regardless of the serum ferritin.

Therapeutic options:

- Ferrous sulfate tabs (325 mg; 65 mg elemental iron) or ferrous sulfate syrup (300 mg; 65 mg elemental iron) – recommended dose is 325 [300] mg twice daily or thrice daily in adults; 3-6 mg/kg/day of elemental iron for Pediatric patients
- Polysaccharide iron complex capsules (150 mg elemental iron) – recommended dose is 150 mg once daily or twice daily for adults; this dosing form is not recommended for Pediatric patients
- Elemental iron oral drops 15 mg elemental iron/mL – recommended dose is 150 mg once or twice daily in adults; 3-6 mg/kg/day for Pediatric patients
- Iron sucrose (VENOFER) Injection:
 - Hemodialysis dependent: 100 mg on HD days (maximum 1000 mg)
 - Non-dialysis dependent: 200 mg x 5 doses over a 14-day period (maximum 1000 mg)
 - In select non-dialysis dependent patients: 500mg x 2 doses (maximum 1000mg) may be administered to reduce number of infusions
 - Pediatric patients: 0.5 mg/kg/dose (max 100 mg) every 2 weeks for 6 doses (if HD dependent) or every 4 weeks for 3 doses (if not HD dependent)

2. Folic Acid:

Adults: 1 mg/day IM, IV, ORAL, SC (resistant cases may require larger doses)

Infants: 15 mcg/kg daily via any route

Children: 1mg per day via any route

3. Vitamin B-12 (cyanocobalamin):

Adults: 200 - 1000 mcg orally daily or 100 mg IM monthly.

Pediatrics: 0.2 mcg/kg IM or SC for 2 days followed by 1000 mcg/day for 2-7 days, followed by 100 mcg per week for a month.

Appendix E: Dose Adjustment Secondary to Changes in Hgb

If Hgb has not increased by 1 g/dL from baseline in 2 weeks, increase dose by 25% (round the dose to the nearest syringe size; maximum allowable dose is 200 mcg) and recheck Hgb in 1 week.

If Hgb response is greater than 1.5 g/dL at 3 weeks or less, reduce dose by 25% (round the dose to the nearest syringe size; maximum allowable dose is 200 mcg) and recheck Hgb in 1 week.

If Hgb is increasing and approaching the Hgb threshold, reduce the dose by 25% (round the dose to the nearest syringe size; maximum allowable dose is 200 mcg) and recheck Hgb in 1 week.

If Hgb is greater than 10 g/dL for anemia associated with low-risk Myelodysplastic Syndrome, or 11.5 g/dL for anemia of chronic kidney disease, hold that day's dose. Reorder for the next standardized administration day if Hgb is above the threshold secondary to RBC transfusion, otherwise maintain current day and schedule for the following week. Decrease the subsequent doses by 25% (round the dose to the nearest syringe size; maximum allowable dose is 200 mcg) unless patient's Hgb is above the threshold secondary to RBC transfusion. Recheck Hgb in 1 week.

If the patient does not respond after 4 months of therapy, ESA therapy should be considered a failure and discontinued.

Appendix F: Contraindications for ESA Therapy

1. Hgb greater than 10 g/dL for anemia associated with low-risk Myelodysplastic Syndrome, and greater than 12 g/dL for all other indications
2. Uncontrolled hypertension
3. Epoetin alfa is contraindicated if patient has a hypersensitivity to albumin. Darbepoetin does not contain albumin and can be safely administered.

Appendix G: Recommendations for the Management of Anemia in Patients Refusing Blood Transfusions

1. Recommend epoetin alfa over darbepoetin alfa secondary to greater volume of literature utilizing the former in this population. Recommend to target a goal hemoglobin of no more than 12 g/dL.
2. Recommendations for preoperative management:
 - a. Initiate therapy one month prior to operation with 600 units/kg subcutaneously per week for 4 doses (with the last dose administered on the day before surgery)
3. Recommendations if no preoperative management ESA utilized
 - a. Initiate epoetin alfa 40,000 units subcutaneously daily until the hemoglobin is greater than or equal to 12 g/dL (if not concurrently on investigational Hemopure) or greater than or equal to 7 g/dL (if concurrently on investigational Hemopure) can then decrease epoetin alfa dose to 40,000 units subcutaneously weekly.
 - i. If a patient refuses albumin products, darbepoetin 100 mcg subcutaneously daily can be substituted for epoetin alfa
 - b. If the patient is less than 60 kg, initiate epoetin alfa as above but at a dose of 500 units/kg/day.
4. Recommend adjunctive therapy to optimize ESA efficacy:
 - a. Iron sucrose 250 mg IV daily x 4 doses and then consider conversion to oral iron
 - b. Ascorbic acid 500 mg PO TID (Decrease dose to 500 mg PO daily if CrCl < 30ml/min)
 - c. Folic acid 1 mg PO daily
 - d. Vitamin B-12 1000 mcg IM daily x 6 doses then start weekly x 5 doses

Appendix H: Pharmacist Monitoring and Documentation

Upon receipt of the initial ESA order the pharmacist will attach an iVent to the order as follows:

- Type should be marked as “Therapeutic Monitoring”
- Subtype should be left blank
- Status should be left “Open”
- Scratch Notes should be used to designate the standardized day of administration
- Use the “UF RX ESA I-VENT NOTE - INITIAL” smart text for the initial documentation

After the initial order is verified, the ESA therapy should be monitored weekly on the standard day of administration (Monday or Thursday) prior to the standard administration time of 1800

- Use the “UF RX ESA I-VENT NOTE – FOLLOW UP” smart text for weekly follow up documentation

Appendix I: Summary of Pharmacist Activities

The following activities can be performed by a pharmacist without prior prescriber approval or verbal order:

- Standardize the dose (as per appendix A)
- Change the day the second dose is administered (as per appendix B)
- Interchange between epoetin alfa and darbepoetin alfa unless prescriber specifies “Do Not Substitute” (as per appendix A)
- Round dose (as per appendix A)
- Change the day of administration if Hgb greater than 10 g/dL for anemia associated with low-risk Myelodysplastic Syndrome, or 11.5 g/dL for anemia of chronic kidney disease.
- Order the following laboratory test (under conditions specified in appendix C): Complete blood count (CBC), folate, hemoglobin, serum ferritin, serum iron, total iron-binding capacity, reticulocyte count, and vitamin B-12
- Change ESA dose secondary to lack of Hgb response or greater than desired increase in Hgb (as per appendix E)
- Write order to not administer the scheduled ESA dose under the following conditions:
 - If iron, folic acid or vitamin B-12 deficiency is identified as source of anemia
 - If Hgb greater than 10 g/dL for anemia associated with low-risk Myelodysplastic Syndrome, or 11.5 g/dL for anemia of chronic kidney disease.
 - If patient has uncontrolled hypertension
- Discontinue epoetin alfa if patient has hypersensitivity to albumin.
 - Recommend use of darbepoetin as this ESA is albumin free.
- Document recommendations and rationale for orders in progress note section of chart and the i-vent linked to that patient or their ESA therapy.

The following activities require collaboration with the prescriber and can be performed by the pharmacist **ONLY** with prescriber authorization or verbal order:

- Order iron supplementation
- Order folic acid supplementation
- Order vitamin B-12 supplementation

References

- KDOQI Clinical Practice Guidelines and Clinical Practice Recommendations for Anemia in Chronic Kidney Disease. *Am J Kidney Dis.* 2012;4(2):S283-287.
- Improving Global Outcomes (KDIGO) Anemia Work Group. KDIGO Clinical Practice Guideline for Anemia in Chronic Kidney Disease. *Kidney Inter.* 2012;2:S279–335.
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- Wise C, Chan R. The management of anemia in a critically ill Jehovah's Witness patient using darbepoetin alfa. *Crit Care Med.* 2020; 48 (1):259
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